

Supplementary Material

Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Pair-Level Study

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This supplement holds material moved from the main paper: the per-model precision analysis that motivates the pair-level design (S1), the record of findings withdrawn from the earlier version of this work (S2), the simulation check and its post-hoc audit (S3), detailed robustness results for every sample (S4), the exploratory analysis of agreement as a signal of shared lineage (S5), the post-hoc analyses added after the third review (S6), and the execution record and data-integrity audit of the earlier per-model evaluation, which is not used for inference (S7); and the lineage-resolution algorithm (S8). Every table and figure is generated from the committed outputs by the repository’s table and figure scripts.

S1. Why per-model variance decomposition is not enough

Model and identification

Let Y_i be a per-model trait (e.g., accuracy), with family $f(i)$ and release quarter $e(i)$:

$$\begin{aligned} Y_i &= \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \\ \alpha &\sim \mathcal{N}(0, \sigma_L^2), \quad \beta \sim \mathcal{N}(0, \sigma_E^2), \quad u \sim \mathcal{N}(0, \sigma_U^2), \end{aligned} \tag{1}$$

so that $\text{Var}(\mathbf{y}) = \sigma_L^2 \mathbf{Z}_F \mathbf{Z}_F^\top + \sigma_E^2 \mathbf{Z}_E \mathbf{Z}_E^\top + \sigma_U^2 \mathbf{I}$, with only the intercept as a fixed effect. The variance components are identified when the covariance basis $\{\mathbf{Z}_F \mathbf{Z}_F^\top, \mathbf{Z}_E \mathbf{Z}_E^\top, \mathbf{I}\}$ is linearly independent. The rank of the fixed-effects dummy design $[\mathbf{1} \mid \mathbf{Z}_F \mid \mathbf{Z}_E]$, used as the gate in the earlier version of this work, is not the relevant condition for this model: for the sixteen models evaluated there, that design is rank-deficient (rank 14 of 15) while the covariance basis has full rank (3 of 3). Its condition number also depends on the arbitrary choice of reference level (TABLE SI).

Precision

Identification does not imply precision. For each design we compute the expected restricted-likelihood information $I_{jk} = \frac{1}{2} \text{tr}(P V_j P V_k)$ at four share scenarios (lineage-dominant 0.5/0.2/0.3, era-dominant 0.2/0.5/0.3, balanced, and low-signal 0.1/0.1/0.8), the delta-method standard error of each share, and, as the verdict, the Monte Carlo root-mean-square error (RMSE) of the share estimates under the same REML estimator (500 replicates per scenario). The analytic standard error approximates the Monte Carlo RMSE (e.g., 0.289 versus 0.285 for the family share of the sixteen-model design in the lineage-dominant scenario) but overstates it in the low-signal scenario (era share: 0.411 versus 0.275), so it is a conservative screen before any measurement and the Monte Carlo RMSE is the verdict.

Table SI: Precision of family and era variance shares for per-model traits. FE: fixed-effects dummy design used as a gate in the earlier version; κ range is over every choice of reference levels. Covariance-basis rank 3/3 means the variance components are identified. The last column is the worst-case Monte Carlo RMSE of either share across four scenarios (500 replicates each).

Design	Families	Quarters	FE full rank	$\kappa(X^\top X)$ range	Cov. basis	Worst RMSE
16 evaluated models	5	11	no	singular	3/3	0.302
22-model selection	6	14	yes	285–1526	3/3	0.256
47-model candidate frame	6	14	yes	166–1116	3/3	0.189
30-model “sufficient” design	6	8	yes	91–144	3/3	0.194
42-model sweep design	7	8	yes	83–113	3/3	0.182
nested reference (72)	6	6	no	singular	2/3	not identified

No design reaches an RMSE of 0.10 for both shares (TABLE SI): not the sixteen evaluated models (0.30), not the 47-model candidate frame (0.19), and not the 30-model configuration the earlier version described as sufficient (0.19). Removing the single model responsible for the fixed-effects rank deficiency leaves the RMSE unchanged (0.302). The binding constraint is the number of *levels* of each factor: more models per level help only up to a point (with six families and fourteen quarters, 22 models give 0.256 and 47 give 0.189). In staggered designs (Fig. S1), bringing both shares within ± 0.10 requires on the order of sixty families and fourteen quarters; with eight quarters the era share never falls below about 0.14. Because release quarters are few by construction, a per-model decomposition is a poor instrument for the paper’s research questions in populations like those examined here. Within the designs and scenarios examined, this motivates the pair-level design; it does not show that the shares are inestimable under every design, only that the required numbers of families and quarters exceed those available.

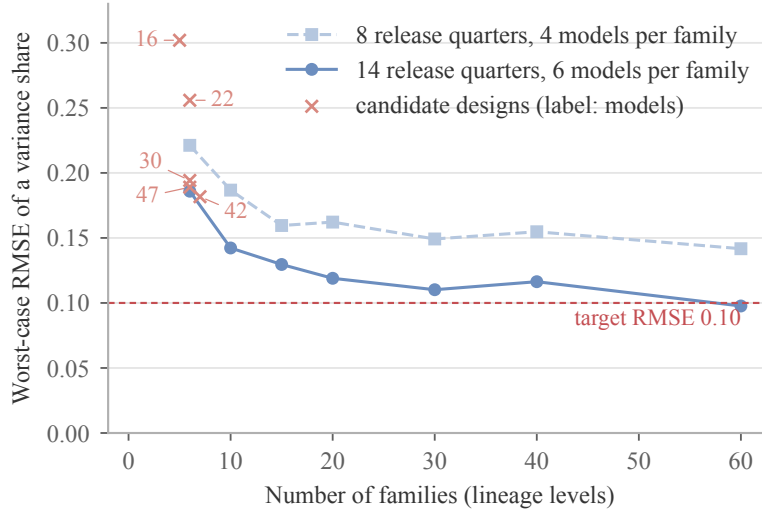


Figure S1: Why per-model variance decomposition is too imprecise. Worst-case Monte Carlo RMSE of the family or era variance share across four scenarios, for staggered designs with 8 or 14 release quarters and for five candidate designs from real populations (crosses, labelled by number of models). The dashed line marks 0.10. RMSE is the error of the estimated share, not a confidence-interval half-width. Traits are simulated on each design’s family and quarter structure; the design labelled 16 takes only the family and quarter labels of the 16 models run in the earlier per-model evaluation (Section S7), not their measured accuracies, and the figure does not validate those measurements.

S2. Changes from the earlier version

The earlier version of this manuscript (archived in the repository as [paper/src/archive/](#)) reported a sixteen-model evaluation and an identifiability gate. TABLE SII lists each finding of that version that is withdrawn or corrected, why, and how it is treated here.

Table SII: Findings of the earlier version and their treatment.

Earlier claim	Reason	Current treatment
Variance components “not identifiable” (fixed-effects dummy-design gate)	Random-effects model identified (basis rank 3/3); condition number depends on reference level (115–585)	Withdrawn; imprecision quantified (Section S1)
Item-level predictions; Qwen-7B accuracy 0.2294	Code read only option A’s log-likelihood; 0.2294 is the all-A rate	Withdrawn; code fixed; validation against stored correctness (main paper, Section III)
30-model design “passes” at $\kappa \leq 50$	Its κ is 93	Corrected
Sweep enforced crossing and connectedness	100 of 105 crossed, 60 connected	Corrected
LPM and GLMM both hit the era boundary in 20–60% of replicates	Only the GLMM; LPM 0–6.7%	Corrected
Nested-design detector: “zero silent coverage”	Undefined: no replicate went undetected	Corrected
Sixteen-model population as pre-specified	Four planned models not run (GPU budget); the DeepSeek imputation was specified but not executed; per-question outputs invalid	Arm not used for inference (Section S7)

S3. Simulation check of separability

The main paper (Section IV-G) describes the simulated data and criteria. Table SIII gives the pre-specified check (20 replicates per scenario) for three caps on the number of models per root; Table SIV gives the post-hoc audit with 500 replicates, shown in Fig. S2. With 20 replicates the rates move in steps of five points, which explains why the audit ranks the caps differently. Cap 40 remains the primary design because it was the dated choice made before any pair outcome was computed.

Table SIII: Pre-specified simulation check (primary roster, 14,042 simulated items, 20 replicates per scenario). Null: largest false-rejection rate. Leakage: largest rejection of the wrong coefficient. Power: rejection of the lineage / same-month coefficient at the stated effect weight.

Cap	N	Null	Leakage	Power $\lambda=0.05$	Power $\lambda=0.03$
15	561	15%	20%	100% / 100%	50% / 75%
40	613	5%	10%	100% / 100%	40% / 95%
none	721	10%	15%	100% / 100%	95% / 95%

Table SIV: Post-hoc gate audit (500 replicates per truth; Wilson 95% intervals, in percent). Null: largest rejection rate of any term under no effect. Leakage: largest cross-term rejection across $\lambda \in \{0.03, 0.05, 0.10\}$. Power: rejection of the lineage (L) or release-month (E) term. Post hoc.

Sample	Cap	N	Null	Leakage	L, $\lambda=0.05$	E, $\lambda=0.05$	L, $\lambda=0.03$	E, $\lambda=0.03$
Primary	15	561	6.0 [4.2, 8.4]	7.2 [5.2, 9.8]	99.6 [98.6, 99.9]	100.0 [99.2, 100.0]	40.2 [36.0, 44.6]	83.8 [80.3, 86.8]
Primary	40	613	7.8 [5.8, 10.5]	10.8 [8.4, 13.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	63.2 [58.9, 67.3]	87.8 [84.6, 90.4]
Primary	none	721	6.6 [4.7, 9.1]	7.2 [5.2, 9.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	90.4 [87.5, 92.7]	90.2 [87.3, 92.5]
Expanded	40	967	5.6 [3.9, 8.0]	12.6 [10.0, 15.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	93.6 [91.1, 95.4]	97.8 [96.1, 98.8]

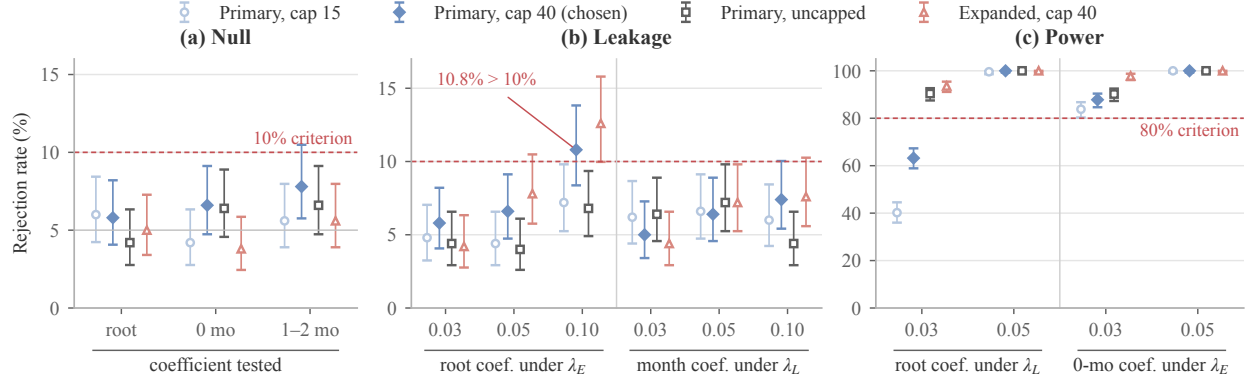


Figure S2: Post-hoc audit of the simulation check (500 replicates per scenario; Wilson 95% intervals). (a) False rejection of each coefficient under no effect. (b) Leakage: rejection of the lineage coefficient under release-time-only effects (λ_E), and the larger rejection rate of the 0- and 1–2-month coefficients under lineage-only effects (λ_L). (c) Power. Upper tick labels give the strength of the simulated effect; lower labels give the coefficient tested and the effect simulated. λ_L : lineage effect (probability of a root-specific wrong option); λ_E : release-time effect (probability of a month-specific wrong option); root: shared-root coefficient; month: the 0- or 1–2-month coefficient, whichever is rejected more often; 0 mo: same-month coefficient. Dashed lines: the pre-specified criteria. The chosen design (primary, cap 40) exceeds the 10% leakage criterion at $\lambda_E = 0.10$ (10.8%).

S4. Detailed robustness results

The main paper’s robustness table reports the primary sample. The tables below give every sample; figures show the same estimates. “Same month” is the 0-month gap bin relative to 12 or more months. S2: models with accuracy ≥ 0.30 .

Table SV: Coefficients with and without the accuracy terms (95% CIs from the delete-one-root jackknife).

Term	Primary		Expanded	
	Without	With	Without	With
Shared root	+0.283 [+0.251, +0.315]	+0.133 [+0.108, +0.158]	+0.240 [+0.205, +0.275]	+0.127 [+0.110, +0.145]
Same month	+0.087 [+0.051, +0.122]	+0.004 [−0.005, +0.013]	+0.092 [+0.065, +0.119]	+0.006 [−0.002, +0.014]
1–2 months	+0.085 [+0.048, +0.123]	0.000 [−0.007, +0.007]	+0.087 [+0.060, +0.114]	+0.002 [−0.005, +0.010]
3–5 months	+0.065 [+0.031, +0.099]	−0.001 [−0.009, +0.007]	+0.078 [+0.050, +0.107]	+0.001 [−0.007, +0.008]
6–11 months	+0.029 [+0.012, +0.046]	−0.001 [−0.008, +0.006]	+0.055 [+0.031, +0.079]	+0.002 [−0.006, +0.009]

Table SVI: Inference procedures. [†]Pre-specified. Root-level procedures treat lineage roots as the unit of dependence.

Sample	Term	Standard error	Estimate [95% CI]	<i>p</i>
Primary	same root	model-dyadic [†]	+0.133 [+0.118, +0.148]	<0.001
Primary	same root	root-dyadic	+0.133 [+0.114, +0.152]	<0.001
Primary	same root	delete-one-root jackknife	+0.133 [+0.108, +0.158]	<0.001
Primary	same month	model-dyadic [†]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	same month	root-dyadic	+0.0039 [−0.0041, +0.0118]	0.343
Primary	same month	delete-one-root jackknife	+0.0039 [−0.0048, +0.0125]	0.383
Primary, S2	same root	model-dyadic [†]	+0.120 [+0.105, +0.135]	<0.001
Primary, S2	same root	root-dyadic	+0.120 [+0.103, +0.136]	<0.001
Primary, S2	same root	delete-one-root jackknife	+0.120 [+0.098, +0.142]	<0.001
Primary, S2	same month	model-dyadic [†]	−0.0145 [−0.0268, −0.0022]	0.021
Primary, S2	same month	root-dyadic	−0.0145 [−0.0269, −0.0021]	0.022
Primary, S2	same month	delete-one-root jackknife	−0.0145 [−0.0303, +0.0013]	0.072
Expanded	same root	model-dyadic [†]	+0.127 [+0.115, +0.140]	<0.001
Expanded	same root	root-dyadic	+0.127 [+0.112, +0.143]	<0.001
Expanded	same root	delete-one-root jackknife	+0.127 [+0.110, +0.145]	<0.001
Expanded	same month	model-dyadic [†]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	same month	root-dyadic	+0.0059 [−0.0017, +0.0135]	0.128
Expanded	same month	delete-one-root jackknife	+0.0059 [−0.0023, +0.0142]	0.160
Expanded, S2	same root	model-dyadic [†]	+0.122 [+0.111, +0.133]	<0.001
Expanded, S2	same root	root-dyadic	+0.122 [+0.108, +0.136]	<0.001
Expanded, S2	same root	delete-one-root jackknife	+0.122 [+0.105, +0.139]	<0.001
Expanded, S2	same month	model-dyadic [†]	−0.0089 [−0.0185, +0.0008]	0.071
Expanded, S2	same month	root-dyadic	−0.0089 [−0.0191, +0.0013]	0.088
Expanded, S2	same month	delete-one-root jackknife	−0.0089 [−0.0218, +0.0040]	0.178

Table SVII: Outcome and weighting variants (95% CIs, model-level dyadic SEs). Every pair has at least 1,166 jointly wrong items, so the variant restricted to pairs with at least 1,000 is identical to the pre-specified fit.

Sample	Variant	Shared root	Same month	<i>p</i> (same month)
Primary	primary outcome, OLS (pre-specified)	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	WLS, weight = jointly-wrong items	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]	0.023
Primary	pairs with ≥ 1000 jointly-wrong items	+0.133 [+0.118, +0.148]	+0.0039 [−0.0029, +0.0106]	0.263
Primary	phi of error indicators (secondary)	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]	<0.001
Primary	agreement minus option-distribution chance (exploratory)	+0.125 [+0.111, +0.140]	−0.0018 [−0.0046, +0.0011]	0.227
Expanded	primary outcome, OLS (pre-specified)	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	WLS, weight = jointly-wrong items	+0.136 [+0.122, +0.151]	+0.0114 [+0.0033, +0.0195]	0.006
Expanded	pairs with ≥ 1000 jointly-wrong items	+0.127 [+0.115, +0.140]	+0.0059 [−0.0003, +0.0121]	0.061
Expanded	phi of error indicators (secondary)	+0.163 [+0.146, +0.180]	+0.0194 [+0.0118, +0.0270]	<0.001
Expanded	agreement minus option-distribution chance (exploratory)	+0.116 [+0.105, +0.127]	+0.0007 [−0.0019, +0.0033]	0.604

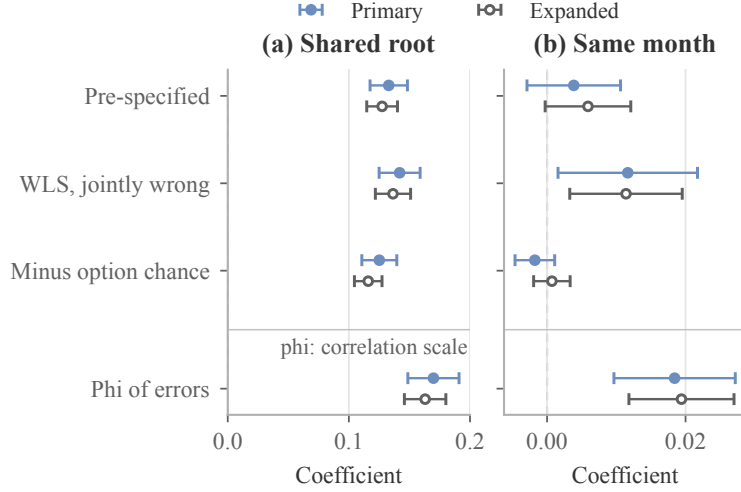


Figure S3: Outcome and weighting variants (95% CIs). Phi is a correlation, on a different scale.

Table SVIII: Item-difficulty-aware nulls (post hoc). Mean: average outcome across pairs. 95% CIs and p -values from the delete-one-root jackknife. N1a: observed agreement minus the item-level distractor null computed from all models; N1b: the same with roots weighted equally; N2: co-failure beyond a Rasch model, per item.

Sample	Outcome	Mean	Shared root	p	Same month	p
Primary	Same-wrong agreement (primary outcome)	0.423	+0.133 [+0.108, +0.158]	<0.001	+0.0039 [-0.0048, +0.0125]	0.383
Primary	N1a: minus item distractor null	0.002	+0.133 [+0.107, +0.159]	<0.001	+0.0040 [-0.0047, +0.0127]	0.367
Primary	N1b: as N1a, roots weighted equally	0.019	+0.133 [+0.108, +0.159]	<0.001	+0.0038 [-0.0049, +0.0124]	0.393
Primary	N2: co-failure over Rasch null (per item)	0.000	+0.040 [+0.032, +0.048]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Primary, S2	Same-wrong agreement (primary outcome)	0.551	+0.120 [+0.098, +0.142]	<0.001	-0.0145 [-0.0303, +0.0013]	0.072
Primary, S2	N1a: minus item distractor null	0.000	+0.119 [+0.096, +0.142]	<0.001	-0.0138 [-0.0310, +0.0034]	0.116
Primary, S2	N1b: as N1a, roots weighted equally	0.023	+0.120 [+0.097, +0.143]	<0.001	-0.0143 [-0.0315, +0.0030]	0.104
Primary, S2	N2: co-failure over Rasch null (per item)	0.000	+0.035 [+0.028, +0.042]	<0.001	+0.0008 [-0.0019, +0.0036]	0.561
Expanded	Same-wrong agreement (primary outcome)	0.434	+0.127 [+0.110, +0.145]	<0.001	+0.0059 [-0.0023, +0.0142]	0.160
Expanded	N1a: minus item distractor null	0.004	+0.128 [+0.110, +0.146]	<0.001	+0.0061 [-0.0024, +0.0146]	0.160
Expanded	N1b: as N1a, roots weighted equally	0.024	+0.128 [+0.110, +0.146]	<0.001	+0.0062 [-0.0023, +0.0146]	0.154
Expanded	N2: co-failure over Rasch null (per item)	0.000	+0.038 [+0.032, +0.045]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Expanded, S2	Same-wrong agreement (primary outcome)	0.547	+0.122 [+0.105, +0.139]	<0.001	-0.0089 [-0.0218, +0.0040]	0.178
Expanded, S2	N1a: minus item distractor null	0.000	+0.122 [+0.105, +0.140]	<0.001	-0.0088 [-0.0226, +0.0051]	0.214
Expanded, S2	N1b: as N1a, roots weighted equally	0.019	+0.123 [+0.106, +0.141]	<0.001	-0.0088 [-0.0229, +0.0052]	0.216
Expanded, S2	N2: co-failure over Rasch null (per item)	0.000	+0.037 [+0.031, +0.043]	<0.001	+0.0016 [-0.0028, +0.0060]	0.480

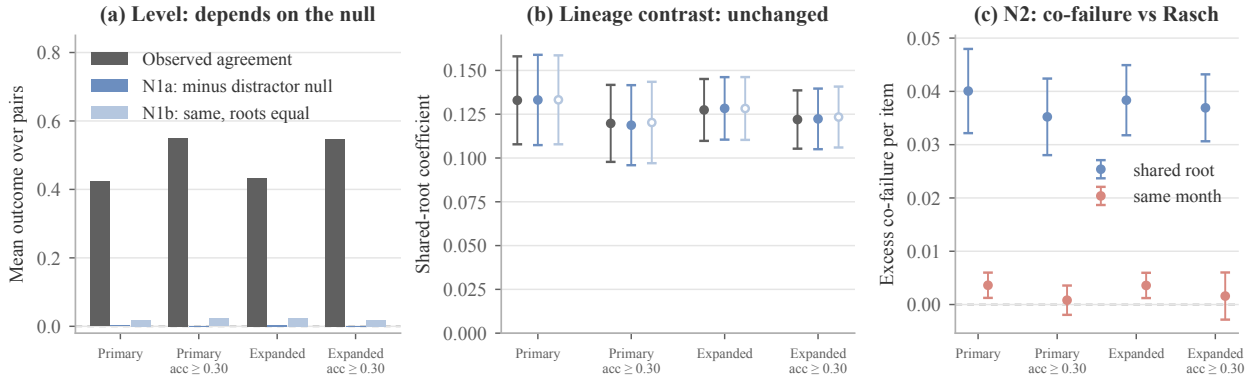


Figure S4: Item-difficulty-aware nulls. (a) Mean outcome over pairs: the level of agreement depends on the null. (b) Shared-root coefficient: the lineage contrast does not. (c) Co-failure beyond a Rasch model.

Table SIX: Size and architecture controls (exploratory; all include the S1 covariate). Entries: estimate (SE).

Sample	Models	Shared root (root-clustered SE)	Shared root (+ size, architecture)	Same month (+ size, architecture)	Same architecture
Primary	590	+0.1328 (0.0096)	+0.1256 (0.0082)	+0.0071 (0.0050)	+0.0027 (0.0026)
Primary, S2	343	+0.1166 (0.0086)	+0.1018 (0.0087)	-0.0205 (0.0110)	+0.0182 (0.0042)
Expanded	928	+0.1263 (0.0072)	+0.1201 (0.0062)	+0.0115 (0.0045)	+0.0050 (0.0022)
Expanded, S2	589	+0.1200 (0.0069)	+0.1085 (0.0057)	-0.0088 (0.0081)	+0.0181 (0.0024)

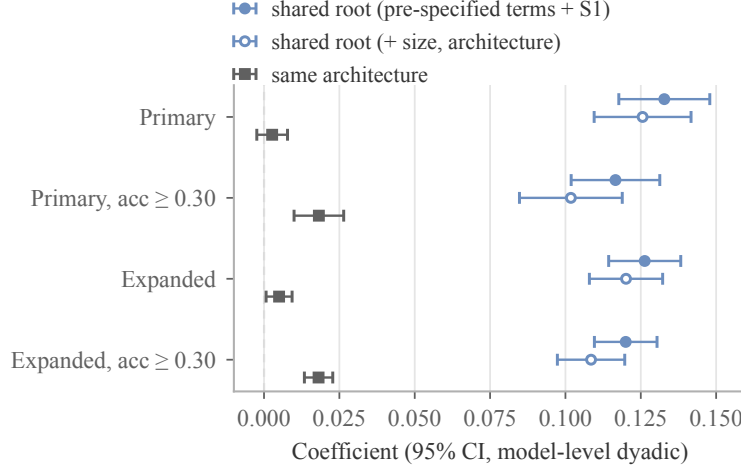


Figure S5: Size and architecture controls (95% CIs, model-level dyadic SEs).

Table SX: Further post-hoc analyses (delete-one-root jackknife 95% CIs). Equivalence bound: largest absolute same-month coefficient not excluded by the 90% interval. Root release month: gap bins from the roots' release months. CAPA: chance-adjusted agreement from chosen options. Clean sample: degenerate models removed and identical-prediction clusters collapsed. Config-consistent: models whose configuration file points to another root removed. Tree distance: primary sample only.

	Primary	Primary, S2	Expanded	Expanded, S2
Same month: 90% equivalence bound	±0.011	±0.028	±0.013	±0.020
Root release month: shared root	+0.131 [+0.108, +0.154]	+0.122 [+0.103, +0.140]	+0.119 [+0.101, +0.136]	+0.112 [+0.096, +0.127]
Root release month: same month	-0.001 [-0.011, +0.010]	-0.022 [-0.040, -0.004]	+0.009 [-0.004, +0.022]	+0.001 [-0.017, +0.019]
CAPA outcome: shared root	+0.194 [+0.153, +0.234]	+0.166 [+0.131, +0.201]	+0.183 [+0.156, +0.210]	+0.169 [+0.144, +0.195]
CAPA outcome: same month	+0.013 [+0.002, +0.023]	-0.002 [-0.013, +0.010]	+0.014 [+0.004, +0.023]	+0.002 [-0.010, +0.014]
Clean sample: shared root	+0.128 [+0.106, +0.150]	+0.117 [+0.096, +0.137]	+0.125 [+0.107, +0.142]	+0.120 [+0.104, +0.135]
Clean sample: same month	-0.002 [-0.008, +0.003]	-0.015 [-0.031, +0.001]	+0.002 [-0.005, +0.009]	-0.009 [-0.022, +0.004]
Config-consistent lineage: shared root	+0.135 [+0.108, +0.161]	-	-	-
Tree distance 1	+0.185 [+0.157, +0.213]	-	-	-
Tree distance 2	+0.136 [+0.117, +0.155]	-	-	-
Tree distance 3+	+0.115 [+0.103, +0.128]	-	-	-
Upload and root month together: same upload month	+0.018 [+0.003, +0.033]	+0.013 [-0.007, +0.033]	+0.007 [-0.005, +0.019]	-0.007 [-0.021, +0.007]
Upload and root month together: same root month	-0.015 [-0.030, -0.001]	-0.033 [-0.058, -0.007]	+0.005 [-0.012, +0.021]	+0.005 [-0.018, +0.027]

S5. Agreement as a signal of shared lineage

This analysis is post hoc and descriptive. If related models share wrong answers, a pair's agreement should carry information about whether the two share a root. In the primary sample, agreement alone separates same-root from different-root pairs with an area under the receiver operating characteristic curve (AUC) of 0.95, and agreement net of the accuracy and release-gap terms of the pre-specified model with 0.88 (Fig. S6). Above an agreement of 0.7 lie 68% of same-root pairs but only 2.1% of different-root pairs. Because different-root pairs outnumber same-root pairs by 85 to 1, however, most pairs above that threshold are still unrelated (about 3,600 against 1,400), so agreement on one benchmark flags candidate relatives rather than establishing them. AUC here is a descriptive summary, not the accuracy of a validated lineage detector.

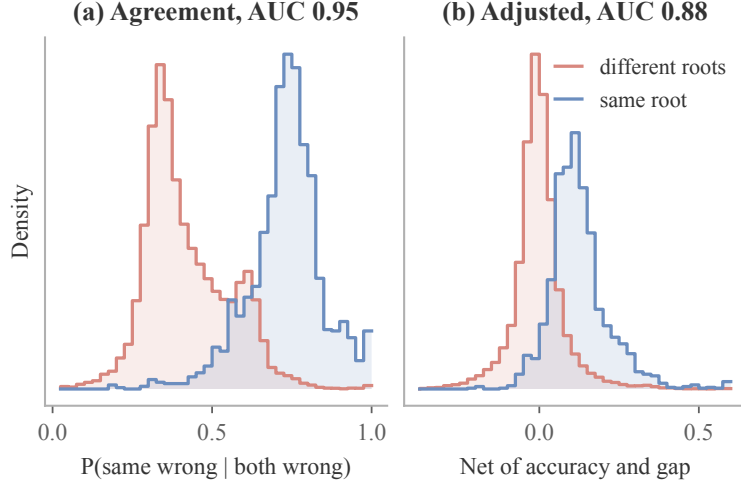


Figure S6: Agreement as a signal of shared lineage (primary sample; 2,020 same-root and 171,735 different-root pairs; densities). (a) Observed agreement on wrong answers. (b) Agreement after removing the accuracy and release-gap terms of the pre-specified model. AUC: probability that a random same-root pair scores higher than a random different-root pair.

S6. Post-hoc analyses from the third review

All analyses in this section were run after the main results were known and were not specified in advance. Code: `scripts/run_exp04_revision3.py`; outputs: `results/exp04_revision3/`.

Flexible accuracy control and answer-key errors

The pre-specified model controls accuracy through the linear sum and absolute difference of the two models' accuracies. TABLE SXI replaces them with a full quadratic surface, with fixed effects for every unordered pair of accuracy bins (deciles: 55 cells; 20 bins: 210 cells), and with the subset of pairs whose accuracies differ by at most 0.02 (plus decile cells). The answer-key check matches 5,698 of the 14,042 items to MMLU-Redux 2.0, which re-annotates 100 items per subject; 369 of the matched items are annotated as erroneous (wrong ground truth 106, no correct answer 36, several correct answers 39, unclear question or options 156, expert knowledge needed 32) and 5,329 as correct. Accuracy is recomputed on each item subset.

Table SXI: Accuracy specification and answer-key errors (post hoc; delete-one-root jackknife 95% CIs).

	Primary	Primary, S2	Expanded	Expanded, S2
<i>Shared root</i>				
Pre-specified (linear accuracy terms)	+0.133 [+0.108, +0.158]	+0.120 [+0.098, +0.142]	+0.127 [+0.110, +0.145]	+0.122 [+0.105, +0.139]
No accuracy terms (total association)	+0.283 [+0.251, +0.315]	+0.185 [+0.152, +0.218]	+0.240 [+0.205, +0.275]	+0.184 [+0.163, +0.206]
Quadratic accuracy surface	+0.123 [+0.094, +0.153]	+0.115 [+0.088, +0.141]	+0.117 [+0.101, +0.134]	+0.118 [+0.100, +0.136]
Accuracy cells: deciles (55 cells)	+0.127 [+0.096, +0.158]	+0.113 [+0.078, +0.148]	+0.125 [+0.104, +0.145]	+0.124 [+0.104, +0.143]
Accuracy cells: 20 bins (210 cells)	+0.119 [+0.084, +0.153]	+0.110 [+0.079, +0.141]	+0.118 [+0.099, +0.137]	+0.117 [+0.098, +0.136]
Accuracy-matched pairs ($ \Delta \leq 0.02$) + decile cells	+0.148 [+0.107, +0.189]	+0.134 [+0.098, +0.169]	+0.149 [+0.116, +0.181]	+0.133 [+0.102, +0.163]
MMLU-Redux: flagged items removed	+0.134 [+0.109, +0.159]	+0.121 [+0.098, +0.143]	+0.128 [+0.110, +0.146]	+0.122 [+0.106, +0.139]
MMLU-Redux items only	+0.131 [+0.105, +0.156]	+0.118 [+0.095, +0.140]	+0.128 [+0.109, +0.147]	+0.122 [+0.105, +0.139]
MMLU-Redux items annotated correct only	+0.133 [+0.107, +0.159]	+0.119 [+0.096, +0.143]	+0.130 [+0.111, +0.149]	+0.124 [+0.107, +0.140]
<i>Same month</i>				
Pre-specified (linear accuracy terms)	+0.004 [-0.005, +0.013]	-0.015 [-0.030, +0.001]	+0.006 [-0.002, +0.014]	-0.009 [-0.022, +0.004]
No accuracy terms (total association)	+0.087 [+0.051, +0.122]	+0.102 [+0.054, +0.150]	+0.092 [+0.065, +0.119]	+0.086 [+0.034, +0.138]
Quadratic accuracy surface	+0.004 [-0.004, +0.013]	-0.012 [-0.028, +0.003]	+0.006 [-0.002, +0.014]	-0.010 [-0.023, +0.004]
Accuracy cells: deciles (55 cells)	0.000 [-0.006, +0.006]	+0.002 [-0.010, +0.015]	+0.001 [-0.005, +0.008]	+0.003 [-0.007, +0.012]
Accuracy cells: 20 bins (210 cells)	-0.004 [-0.010, +0.002]	-0.012 [-0.026, +0.001]	-0.001 [-0.007, +0.005]	-0.008 [-0.020, +0.003]
Accuracy-matched pairs ($ \Delta \leq 0.02$) + decile cells	+0.033 [+0.014, +0.052]	+0.034 [-0.006, +0.074]	+0.036 [+0.011, +0.060]	+0.056 [+0.023, +0.089]
MMLU-Redux: flagged items removed	+0.004 [-0.005, +0.013]	-0.015 [-0.031, +0.001]	+0.006 [-0.002, +0.014]	-0.009 [-0.022, +0.004]
MMLU-Redux items only	+0.004 [-0.004, +0.013]	-0.014 [-0.031, +0.003]	+0.007 [-0.002, +0.015]	-0.010 [-0.023, +0.004]
MMLU-Redux items annotated correct only	+0.005 [-0.004, +0.013]	-0.015 [-0.032, +0.002]	+0.007 [-0.002, +0.015]	-0.010 [-0.024, +0.004]

Item fixed effects, bootstrap and permutation

The pair-level outcome conditions on items both models get wrong, so a pair of strong models is compared on harder items than a pair of weak models. The pair $\times$ item model regresses the indicator of the same wrong option on the pair covariates over every (pair, item) cell in which both models are wrong, with a fixed effect for each item; it is computed exactly from item-level aggregates and weights each pair by its number of jointly wrong items, so it is shown beside a weighted pair-level fit without item effects. The root cluster bootstrap resamples lineage roots with replacement (999 draws) and keeps every pair of distinct models among the drawn copies, so a pair from a root drawn k times enters k^2 times and remains a same-root pair, and a pair spanning roots drawn k and m times enters km times (the vertex, or pigeonhole, bootstrap for dyadic data applied to roots; the resampling is checked against explicit enumeration in the test suite). The permutation test shuffles root labels among the models uploaded in the same quarter, preserving root sizes and the time structure (999 permutations, so the smallest attainable p is 0.001). Across the 999 permutations the largest absolute shared-root coefficient is 1.2 points in the primary sample and 1.4 in the expanded sample. The bootstrap interval is asymmetric because resampling roots sometimes omits the two largest roots, whose coefficients are among the smallest.

Table SXII: Item fixed effects and resampling inference (post hoc; 95% CIs).

	CI	Primary		Expanded	
		Shared root	Same month	Shared root	Same month
Item fixed effects	jackknife	+0.142 [+0.107, +0.176]	+0.011 [−0.001, +0.024]	+0.136 [+0.116, +0.157]	+0.011 [+0.001, +0.022]
Same weights, no item effects	jackknife	+0.142 [+0.108, +0.176]	+0.012 [−0.001, +0.024]	+0.136 [+0.116, +0.157]	+0.011 [+0.001, +0.022]
Root cluster bootstrap	bootstrap	+0.133 [+0.114, +0.196]	+0.004 [−0.003, +0.014]	+0.127 [+0.106, +0.162]	+0.006 [−0.002, +0.015]
Root-label permutation	permutation	$p = 0.001$	—	$p = 0.001$	—

Dose-response within roots, and every root counted once

Parent–child pairs come disproportionately from small roots (82 of 177), and small roots show larger coefficients, so the tree-distance gradient could reflect root composition. The within-root model uses same-root pairs only, with a fixed effect for each root, so each distance class is compared with distance 3+ pairs of the same root. The pre-specified coefficient weights roots by their number of same-root pairs, so the two largest roots dominate it; the last two rows of TABLE SXIII count roots instead.

Table SXIII: Within-root dose-response and per-root pooling (primary sample; post hoc). Within-root rows: delete-one-root jackknife 95% CIs over the roots contributing same-root pairs. Random-effects pooling uses DerSimonian–Laird weights with model-level dyadic SEs and is restricted to roots with at least 10 same-root pairs, because a root with a single same-root pair fits it exactly and has no usable SE. Unweighted mean: SE from the spread across roots.

	Estimate [95% CI]	
Within root: tree distance 1 (vs 3+)	+0.055 [+0.041, +0.068]	51 roots
Within root: tree distance 2 (vs 3+)	+0.023 [+0.003, +0.042]	51 roots
Within root: ancestor–descendant (vs more distant)	+0.047 [+0.030, +0.063]	51 roots
Within root: siblings (vs more distant)	+0.022 [+0.002, +0.041]	51 roots
Per-root coefficients, random-effects pooled (≥ 10 same-root pairs)	+0.166 [+0.125, +0.208]	16 roots; $\tau = 0.079$, $I^2 = 0.94$
Per-root coefficients, unweighted mean	+0.253 [+0.211, +0.295]	63 roots

Simulation check with the real accuracies

The original check drew abilities independently of root and month. Here each model’s ability is set so that its expected simulated accuracy equals its observed accuracy, so ability clusters by root and month exactly as in the data; the roster is the analysed primary sample (590 models), with 1,500 items and 200 replicates per scenario. A further scenario has no lineage or release-time effect, but the probability of choosing the common distractor rises with ability ($0.05 + 0.5 \text{acc}^2$), so agreement depends non-linearly on accuracy. Each replicate is fitted with the pre-specified model and with decile accuracy cells.

Table SXIV: Rejection rates (%) with real accuracies (Wilson 95% intervals; model-level dyadic SEs; nominal 5%).

Scenario	Pre-specified accuracy terms		Decile accuracy cells	
	Shared root	Same month	Shared root	Same month
No effect	5.0 [2.7, 9.0]	8.5 [5.4, 13.2]	5.0 [2.7, 9.0]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.03$	6.5 [3.8, 10.8]	28.0 [22.2, 34.6]	5.5 [3.1, 9.6]	27.5 [21.8, 34.1]
Lineage $\lambda_L = 0.03$	14.5 [10.3, 20.0]	7.5 [4.6, 12.0]	15.0 [10.7, 20.6]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.05$	5.0 [2.7, 9.0]	96.0 [92.3, 98.0]	4.5 [2.4, 8.3]	97.5 [94.3, 98.9]
Lineage $\lambda_L = 0.05$	52.5 [45.6, 59.3]	6.0 [3.5, 10.2]	52.0 [45.1, 58.8]	7.0 [4.2, 11.4]
Release-time $\lambda_E = 0.10$	6.0 [3.5, 10.2]	100.0 [98.1, 100.0]	7.5 [4.6, 12.0]	100.0 [98.1, 100.0]
Lineage $\lambda_L = 0.10$	100.0 [98.1, 100.0]	6.0 [3.5, 10.2]	100.0 [98.1, 100.0]	5.0 [2.7, 9.0]
Ability-dependent distractor, no effect	27.0 [21.3, 33.5]	7.0 [4.2, 11.4]	9.0 [5.8, 13.8]	16.0 [11.6, 21.7]

S7. Earlier per-model evaluation: model roster, execution environment and data-integrity audit

This section documents an evaluation that preceded the pair-level design. It is an audit record, not a second empirical study, and it does not validate or support any result of the main paper. None of its measurements enters any result of the main paper or of Sections S3–S6. Section S1 uses only the family and quarter labels of its sixteen recorded models to define one simulated design (the point labelled 16 in Fig. S1). The Appendix of the main paper summarizes this section in two tables: the planned roster with the execution and data status of each entry, and the execution environment with the evidence (recorded, planned, author-reported or unknown) for each item.

Selection and plan

A 47-model candidate frame was reduced to a 22-model roster by a design gate that used only family $\times$ quarter occupancy, declared lineage edges, identifiability constraints and estimated cost, never measured traits (`datasets/coverage/minimum_valid_population.csv`, `g3_report.md`; committed 3 August 2026, last revised 5 August). The two DeepSeek models of the roster (671B and 685B mixture-of-experts) did not fit the available compute; three re-gate variants that excluded them were infeasible because the gate then required other models of the same size class, and a rule to complete their traits by model-based multiple imputation (five draws, seed 2026) was recorded on 16 August 2026, before the first evaluation. The run plan, recorded the same day and also before the first evaluation, assigned the remaining 20 models to one rented GPU. TABLE SXV lists the planned roster. Earlier drafts of the runbook considered an eight-GPU H200 node and a split across several machines; neither was used.

Table SXV: Planned roster of the earlier per-model evaluation (datasets/coverage/trait_definition.csv). Size: as listed in the plan. Gated: the repository requires an accepted licence and a Hugging Face token.

Model	Family	Quarter	Size	Planned repository	Planned precision	Gated
Llama-1	Llama	2023Q1	7B	meta-llama/Llama-1	bf16	yes
Phi-1	Phi	2023Q2	1.3B	microsoft/phi-1	bf16	no
Mistral-7B	Mistral	2023Q3	7B	mistralai/Mistral-7B-Instruct-v0.3	bf16	no
Phi-1.5	Phi	2023Q3	1.3B	microsoft/phi-1.5	bf16	no
Qwen-7B	Qwen	2023Q3	7B	Qwen/Qwen-7B	bf16	no
Phi-2	Phi	2023Q4	2.7B	microsoft/phi-2	bf16	no
Qwen1.5	Qwen	2024Q1	72B	Qwen/Qwen1.5-72B-Chat	4-bit NF4	no
Phi-3	Phi	2024Q2	14B	microsoft/Phi-3-medium-4k-instruct	bf16	no
Llama-3.1	Llama	2024Q3	70B	meta-llama/Meta-Llama-3.1-70B-Instruct	4-bit NF4	yes
Llama-3.3	Llama	2024Q4	70B	meta-llama/Llama-3.3-70B-Instruct	4-bit NF4	yes
Phi-4	Phi	2024Q4	3.8B	microsoft/Phi-4-mini-instruct	bf16	no
Mistral-Small-3	Mistral	2025Q1	24B	mistralai/Mistral-Small-3-24B-Instruct-2501	bf16	yes
Mistral-Small-3.1	Mistral	2025Q1	24B	mistralai/Mistral-Small-3.1-24B-Instruct-2503	bf16	no
Gemma-3n	Gemma	2025Q2	4B	google/gemma-3n-4b-it	bf16	yes
Mistral-Small-3.2	Mistral	2025Q2	24B	mistralai/Mistral-Small-3.2-24B-Instruct-2510	bf16	yes
Phi-4-reasoning-plus	Phi	2025Q2	11B	microsoft/Phi-4-reasoning-plus	bf16	no
DeepSeek-V3.1	DeepSeek	2025Q3	671B/37B	deepseek-ai/DeepSeek-V3.1	imputed	no
DeepSeek-V3.2	DeepSeek	2025Q4	685B/37B	deepseek-ai/DeepSeek-V3.2	imputed	no
Devstral-2	Mistral	2025Q4	24B	mistralai/Devstral-Small-2509	bf16	yes
Mistral-Small-4	Mistral	2026Q1	32B	mistralai/Mistral-Small-4-32B-Instruct-2603	bf16	yes
Phi-4-reasoning-vision-15B	Phi	2026Q1	15B	microsoft/Phi-4-reasoning-vision-15B	bf16	no
Gemma-4-12B	Gemma	2026Q2	12B	google/gemma-4-12b-it	bf16	no

Execution configuration

The planned configuration (docs/05.Experiments/Exp02_GPU_Runbook.md, addendum of 16 August 2026) was:

- *Hardware.* One RunPod pod with one NVIDIA A100 SXM4 80 GB GPU, not a cluster. The authors report a host with 16 vCPUs and 117 GB of RAM; no pod record in the repository confirms it.
- *Software.* Python 3.11 PyTorch template (PyTorch 2.8.0 as reported by the authors; not recorded); `lm-eval` 0.4.12; `transformers` 4.57.6 (pinned because version 5 removed the `load_in_4bit` argument that `lm-eval` 0.4.12 passes); `accelerate` and `bitsandbytes` (versions not pinned or recorded).
- *Task.* MMLU, 5-shot, log-likelihood scoring of the four answer options (no generation), on the same 14,042 items for every model.
- *Model arguments.* `--dtype bfloat16`, `--attn sdpa`, `--batch auto`; `--quant none` for the 17 models planned in `bfloat16` and `--quant 4bit` for the three 70B-class models; `max.length=2048, truncation=True` for Phi-1, Phi-1.5 and Phi-2; `trust_remote_code` where the repository requires it. The code's 4-bit path passes only `load_in_4bit=True`, which selects the `bitsandbytes` default 4-bit type (FP4), not the NF4 type named in the plan. A loader patch in force during the runs removed quantization arguments for one Mistral model class, so a run tagged 4-bit may not have been quantized as tagged.
- *Procedure.* One model per invocation; check that 14,042 samples were scored; commit the result row and the per-question file; delete the model cache, so that the committed files are the durable record if the pod is reclaimed; previously completed models are skipped on resumption. A pilot (Phi-2, 100 items) failed before any measurement and wrote no data.
- *Intake check.* `lineage_era.phase2_eval_check` with the 20-model manifest (datasets/coverage/a100_full_subset.csv) was to pass before any analysis. It requires one row per planned model, accuracy values in range, per-question row counts equal to the scored samples, valid correctness flags, a common item set, and, since the October 2026 fix, more than one log-likelihood per item and non-constant predictions. No passing run is recorded. Run on the committed outputs on 10 October 2026 it fails with 34 errors: four planned models missing; `acc_norm`

empty for all sixteen rows; and, for every one of the sixteen per-question files, a constant prediction and a single log-likelihood per item. Row counts (14,042), correctness flags and the common item set pass.

Resource provenance

The plan specified a GPU disk of at least 300 GB (the 70B models were to be downloaded in bfloat16 before 4-bit loading, about 140 GB each), about 604 GPU-minutes (about 10 hours) and an estimated cost of USD 16–20 at about USD 1.60 per hour. These are planned values. The commits of results span 16 August 18:08 to 18 August 09:12 UTC; that is wall-clock time including idle periods, not measured run time. No pod configuration, storage record, event log or invoice is in the repository, so the attached disk, run time and cost actually incurred are not reported.

What was recorded

Between 16 and 18 August 2026, results for 16 models were committed, one model per commit (`datasets/phase2_eval_results.csv`, `datasets/eval_samples/`). TABLE SXVI compares them with the plan. These are recorded runs, not validated measurements.

- Qwen1.5-72B, Llama-3.1-70B, Llama-3.3-70B and Phi-4-reasoning-vision-15B were not run: the authors stopped the evaluation after 16 models because further GPU time could not be funded. No decision-log entry records the stop. None of the three models planned in 4-bit was therefore run.
- Mistral-Small-4, planned as a 32B model in bfloat16, was recorded from the repository `Mistral-Small-4-119B-2603` with a 4-bit tag. No decision-log entry records the change; the reproducibility checklist of the earlier version notes only that the 119B model does not fit in bfloat16 on an 80 GB GPU. Five further models were recorded from repositories other than those in the plan (TABLE SXVI).
- *Integrity of the per-question files.* Every per-question file stores one log-likelihood per item and predicts option A on all 14,042 items, because the extraction code read only the first option’s score (`resps[0]`) instead of one score per option. The error was found and fixed in the code on 9 October 2026. No raw `lm-eval` output with all four scores per item was found in the repository, its full history (including stashes and unreachable commits) or the computer used for the analysis, so the per-question predictions cannot be reconstructed.
- *Aggregate accuracies.* The recorded accuracies are `lm-eval`’s own aggregates and do not pass through the faulty extraction, but they cannot be validated item by item. Qwen-7B’s value equals the share of items whose correct answer is A; the leaderboard’s item-validated predictions give 0.584 for the same repository. Five other models score at chance level (0.23–0.25) for reasons not established, including three 24B instruction models. Mistral-Small-3.1 and 3.2 were re-run after their first results, following a loader change that registered their architecture (Mistral3) with the causal-language-model loader (commit `cca711c`; an earlier fix to quantization arguments and rotary-position parameters, `27886ef`, preceded both runs), and remained at chance (0.249 and 0.250 before, 0.234 and 0.231 after). Phi-1.5, the only other model also in the validated leaderboard data, agrees with it (0.422 against 0.424).
- *Imputation.* The DeepSeek imputation was implemented but never executed on measured data, so the planned 22-entry roster was never completed.

No recorded measurement is therefore used. Completing this evaluation, which the authors do not plan to do, would require re-running all twenty planned models with the corrected extraction and resolving the chance-level results.

Table SXVI: Recorded runs of the earlier per-model evaluation against the plan (not used for inference). Recorded repository and precision tag of each committed run, and **lm-eval**’s aggregate MMLU accuracy. Rows, log-lik./item: rows in the per-question file and log-likelihoods stored per item (four are needed). Leaderboard: item-validated accuracy of the same repository in the leaderboard data of the main paper. ^aRepository differs from the plan (TABLE SXV). ^bTag written by the run script; the precision actually used is not confirmed (see Execution configuration).

Model	Recorded repository	Precision	Accuracy	Status
Llama-1	<code>huggyllama/llama-7b^a</code>	bf16	0.342	14,042 rows, 1 log-lik./item
Phi-1	<code>microsoft/phi-1</code>	bf16	0.248	14,042 rows, 1 log-lik./item; chance level, unresolved
Mistral-7B	<code>mistralai/Mistral-7B-Instruct-v0.3</code>	bf16	0.619	14,042 rows, 1 log-lik./item
Phi-1.5	<code>microsoft/phi-1.5</code>	bf16	0.422	14,042 rows, 1 log-lik./item; leaderboard 0.424
Qwen-7B	<code>Qwen/Qwen-7B</code>	bf16	0.229	14,042 rows, 1 log-lik./item; equals the all-A rate; leaderboard 0.584
Phi-2	<code>microsoft/phi-2</code>	bf16	0.564	14,042 rows, 1 log-lik./item
Qwen1.5	–	–	–	not run (evaluation stopped after 16 models: GPU budget)
Phi-3	<code>microsoft/Phi-3-medium-4k-instruct</code>	bf16	0.780	14,042 rows, 1 log-lik./item
Llama-3.1	–	–	–	not run (evaluation stopped after 16 models: GPU budget)
Llama-3.3	–	–	–	not run (evaluation stopped after 16 models: GPU budget)
Phi-4	<code>microsoft/Phi-4-mini-instruct</code>	bf16	0.686	14,042 rows, 1 log-lik./item
Mistral-Small-3	<code>mistralai/Mistral-Small-24B-Instruct-2501^a</code>	bf16	0.807	14,042 rows, 1 log-lik./item
Mistral-Small-3.1	<code>mistralai/Mistral-Small-3.1-24B-Instruct-2503</code>	bf16	0.234	14,042 rows, 1 log-lik./item; chance level, unresolved
Gemma-3n	<code>google/gemma-3n-E4B-it^a</code>	bf16	0.636	14,042 rows, 1 log-lik./item
Mistral-Small-3.2	<code>mistralai/Mistral-Small-3.2-24B-Instruct-2506^a</code>	bf16	0.231	14,042 rows, 1 log-lik./item; chance level, unresolved
Phi-4-reasoning-plus	<code>microsoft/Phi-4-reasoning-plus</code>	bf16	0.778	14,042 rows, 1 log-lik./item
DeepSeek-V3.1	–	–	–	imputation specified, not executed
DeepSeek-V3.2	–	–	–	imputation specified, not executed
Devstral-2	<code>mistralai/Devstral-Small-2-24B-Instruct-2512^a</code>	bf16	0.251	14,042 rows, 1 log-lik./item; chance level, unresolved
Mistral-Small-4	<code>mistralai/Mistral-Small-4-119B-2603^a</code>	4-bit tag ^b	0.243	14,042 rows, 1 log-lik./item; chance level, unresolved
Phi-4-reasoning-vision-15B	–	–	–	not run (evaluation stopped after 16 models: GPU budget)
Gemma-4-12B	<code>google/gemma-4-12b-it</code>	bf16	0.440	14,042 rows, 1 log-lik./item

S8. Lineage-resolution algorithm

Algorithm S1 gives the exact rules used to resolve each model’s declared parent chain to a lineage root (main paper, Section III-B).

Algorithm S1 Lineage root of a model

```

1:  $n \leftarrow$  canonical Hub identifier of the model (renames collapsed);  $d \leftarrow 0$ 
2: while  $d < 20$ :
3:   if  $n$  declares several parents: return unresolved (merge)
4:    $p \leftarrow$  declared parent of  $n$  (expanded population only: at  $d = 0$ , the configuration link if the card declares none)
5:   if  $d > 0$  and  $n$  no longer exists: return unresolved (missing ancestor)
6:   if  $p$  is none and  $n$  is a listed re-upload:  $p \leftarrow$  the checkpoint it copies
7:   if  $p$  is none: if the leaderboard lists  $n$  as a fine-tune, return unresolved (undeclared parent); else return root  $n$ 
8:   if  $p$  was already visited: return unresolved (cycle)
9:    $n \leftarrow$  canonical identifier of  $p$ ;  $d \leftarrow d + 1$ 
10: return unresolved (chain too deep)

```
